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Companion to *A Topological Tidal-Curvature Imager* — Draft for comment

A Placement Law for Shift Sensing in Curved-Spacetime Su-Schrieffer-Heeger Chains

Draft for comment — not peer-reviewed. Companion to *A Topological Tidal-Curvature Imager* (tidal-imager-draft), whose curvature placement law this result extends to stationary metrics. Every claim is DERIVED, COMPUTED, or absent.

Authorship note. This work was developed in extended collaboration between B. Watson (direction, framing, verification protocol) and AI systems (Claude, Anthropic; two independent sessions), which performed the formal derivation, numerics, and drafting under the author's direction. The two-session cross-verification process is described in Appendix B.

Abstract

The curved-spacetime Su-Schrieffer-Heeger (SSH) chain encodes a static diagonal metric in its bond profile and reads it through a topological edge mode; its curvature sensitivity obeys a placement law derived in companion work. Stationary metrics — the Alcubierre class — additionally carry a shift vector g_tx, and the lattice term that encodes a shift is a current-like same-sublattice hopping, which breaks the chiral symmetry protecting the readout. We show this costs nothing where it matters: a particle-hole symmetry (sublattice parity composed with complex conjugation; symmetry class D) survives for arbitrary shift profiles, pinning the edge mode at zero energy to machine precision even under bond disorder, while the naive real form of the same coupling unpins the mode at first order. We then derive the shift placement law: the logarithmic response of the edge density of states equals minus the leaked probability, $d \ln \rho = -\tilde{w}^2 |\psi_0|^2$, where $\tilde{w}$ is the shift in local-hopping units and $|\psi_0|^2$ the mode's probability weight at the shifted bond. The derivation shows the entire disturbance exits through exactly one site of the unprotected sublattice, so the response kernel is diagonal for shift on the mode's own sublattice; for the physical two-sublattice encoding an adjacent-bond cross term of the same order appears, a 5% correction on the warped chain and exactly zero on the flat one. Corollaries: the observable is exactly even in the shift — static calorimetry reads $|g_{tx}|^2$ and is structurally blind to its sign; curvature bias preamplifies shift response, by a factor that grows as $N^2\sigma$ with chain length because the horizon-end bonds vanish as $N^{-\sigma}$ (85× at 100 cells for the profile studied); and a synthetic warp wall of two-percent shift amplitude parked on one element is detectable at SNR ≈ 16–20 in 100 s with demonstrated cryogenic bolometry. An array separates the curvature and shift channels if and only if neighboring elements differ in curvature bias; shift-bias diversity does not work. All results reproduced by two independent implementations in different hopping conventions.

1. Introduction

The companion white paper develops a curvature imager: a dimerized chain whose position-dependent hopping encodes a static diagonal metric, probed by a weakly coupled qubit whose decoherence power reports the edge mode's local density of states, governed by a placement law — sensitivity equals the differential metric tilt integrated over the mode's probability footprint. That instrument's encoding dictionary, inherited from the curved-spacetime SSH construction of Rajbongshi and Modak, covers static diagonal metrics only. The metrics of principal interest to the analog-warp community are stationary: their content lives in a shift vector g_tx, and a sensing program aimed at engineered warp-class geometries must answer a prior question — can a shift be written into the lattice at all without destroying the topological protection the readout is built on?

Tilted Dirac and Weyl cones as lattice realizations of metrics with a shift are established: Könye et al. [5] build tight-binding chains whose position-dependent tilt reproduces horizon physics, and the tilt–shift identification runs through the type-II Weyl literature. What that work does not consider is a protected edge mode on such a chain and what the tilt does to it; that is the question here. The concern is concrete. The lattice route to a tilted cone — the dispersion signature of a shift — runs through same-sublattice hopping, and any same-sublattice term breaks the chiral symmetry that pins the edge mode at zero energy and gives the calorimetric readout its contrast. If protection and encodability draw from the same account, the shift-sensing program is structurally self-defeating. This paper computes and then derives the answer: they do not. The correct encoding preserves a particle-hole symmetry that does the same pinning work; the mode's shape, unlike its energy, responds to the shift; and that response obeys a placement law as clean as the curvature one, with a one-line closed form. The instrument class that emerges is a magnitude detector for synthetic frame-dragging, with its blind spots derived rather than discovered.

2. Encoding a shift, and what protects the mode

Two same-sublattice couplings present themselves. The real next-nearest-neighbor hopping t' contributes a k-even identity term to the dispersion — an energy offset, not a tilt — and it is a protection catastrophe: computed on the warped chain, the zero mode's energy shifts linearly, $E_0 = -0.73\, t'$ (a bond-ratio coefficient, identical in both conventions), confirmed over four decades. The current-like imaginary coupling, $iw(c^\dagger_{n+2}c_n - \text{h.c.})$, contributes $2w \sin k$ — the odd-in-k identity term that tilts the cone and is the lattice form of the shift. Computed on the same chain: the zero mode stays at machine zero ($|E_0| \le 10^{-16}$) to $w = 0.1$, including under one-percent bond disorder across 200 realizations, where the real form at $t' = 0.02$ is unpinned to $|E_0| \approx 1.4 \times 10^{-2}$.

The mechanism is a symmetry handover. The unitary chiral operator S (sublattice parity) no longer anticommutes with H once any same-sublattice term is present, and time reversal is broken by the imaginary hopping. But the antiunitary operator $S\circ K$ — sublattice parity composed with complex conjugation — still anticommutes with H: K flips the sign of the imaginary coupling while S flips the sign of the real nearest-neighbor bonds, so $(S\circ K)H(S\circ K)^{-1} = -H$ exactly, for any shift profile $w(x)$. An antiunitary operator squaring to +1 that anticommutes with H is a particle-hole symmetry: the shifted chain is a one-dimensional class D system with a $\mathbb{Z}_2$ invariant, and the unpaired zero mode of the odd chain is that $\mathbb{Z}_2$ statement — the same protection as a Kitaev-chain Majorana mode. The spectrum remains rigorously $\pm E$ paired (verified below 10^{-14} at $w = 0.1$ with disorder, i.e. at the floor of double precision for a 201-site matrix), and the unpaired mode of an odd chain is pinned at zero. The vulnerability ledger is unchanged by the shift term: on-site potentials and real same-sublattice hopping remain the enemies of the protection; the shift adds no new class.

3. The shift placement law

The pinned mode reshapes under the shift, and the edge observable tracks the reshaping. Write the chiral chain in sublattice blocks, $H_0 = [[0, h],[h^\dagger, 0]]$, with the zero mode $|0\rangle = (a, 0)$ real, normalized, purely on sublattice A, obeying the product structure $a_{m+1}/a_m = -t_1(m)/t_2(m)$. Perturb with one shift bond of strength w on the A–A link between cells m and m+1. Four steps, each verified numerically to the precision of the method:

First, the first-order correction leaks entirely into sublattice B: the perturbation applied to $|0\rangle$ is A-supported with vanishing diagonal element, and the chiral pairing of the excited states cancels its A-components between $\pm E$ partners while their B-components add. The edge site is an A site, so the reading is untouched at first order — the evenness of the observable in w is derived here, not assumed.

Second, the energy stays pinned: the first-order shift vanishes by antisymmetry and the

second-order shift vanishes because the perturbation is A-only while the first-order correction is B-only. This is the perturbative face of the S∘K pinning.

Third — the structural surprise — the leak occupies exactly one site. Solving h ψ_B = −(V|0⟩)_A site by site from the closed end, every B amplitude vanishes until cell m, where ψ_B(m) = iw $a_{m+1}/t_1(m)$; at cell m+1 the two contributions cancel *identically by the zero-mode ratio*, and everything beyond stays zero. The same product structure that builds the mode confines its leak. Verified: one spike, predicted site, predicted amplitude to five significant figures, all other B amplitudes at 10^{-16}.

Fourth, for shift on A–A bonds the second-order change to the A-sublattice shape is a pure renormalization — the A-only perturbation annihilates the B-only first-order correction, so the O(w^2) term is proportional to the zero mode itself, fixed by normalization. The edge reading therefore drops by exactly the leaked probability:

> **$d \ln \rho_{edge} = -\|\psi^{(1)}\|^2 = -w^2\, a_{m+1}^2/t_1(m)^2 = -w^2\, a_m^2/t_2(m)^2$**

The two faces are equal by the mode ratio; in the canonical dimensionless variable $\tilde{w} = w/t_2(m)$ the law is pure footprint, $d \ln \rho = -\tilde{w}^2\, a_m^2$. In words: **the shift sensitivity of the edge reading equals the squared local shift-to-hopping ratio weighted by the mode's own probability footprint — sensitivity is leaked probability.** This is the exact stationary-channel analog of the curvature placement law: same footprint structure, quadratic rather than linear, with the local ratio replacing the accumulated tilt.

The physical encoding, 2w sin k times the identity, places the shift on both sublattices, and the B–B part acts on the leaked amplitude, so the fourth step acquires a cross term of the same order. The second-order A correction then solves h†$\alpha^{(2)}$ = −V_BB $\psi^{(1)}$_B, whose solution is a linear tilt of the A envelope: on the flat chain with uniform w, $\alpha_n/a_n = 1 + \varphi_1 - (n-1)\cdot w^2(1-r^2)/t_1^2$, verified to four digits at two dimerizations. The boundary term obeys $2\varphi_1 = -\|\psi_B^{(1)}\|^2$, and on the flat chain this makes the tilt cancel exactly against the normalization in the edge reading, so the uniform-shift response equals the A–A-only law to 10^{-5} — a cancellation, not diagonality, and the reason every flat-chain check in this program passed. On the warped chain the cancellation fails: the two-sublattice response is 0.947 of the A–A-only law at σ = 0.5, independent of w as w → 0, i.e. a 5% correction at the same order as the law itself. The cross term keeps the footprint structure (the tilt is weighted by the same a_n) and is confined to the bond adjacent to each shifted A–A bond, which is what the counterpart implementation's kernel diagnostics show. An earlier version of this work attributed the 3% gap in a uniform-shift closure to a quartic term; it is this cross term.

4. Corollaries and channel structure

Sign blindness is structural. Complex conjugation maps H(w) to H(−w) with an identical spectrum, so every odd-order response vanishes identically: static calorimetry measures $|g_{tx}|^2$, the magnitude of the frame-dragging, and cannot measure its direction. Reading the sign requires a time-odd observable — directional transport — which is a different instrument, not a refinement of this one.

For shift restricted to A–A bonds the response kernel is exactly diagonal, because each bond's leak occupies a distinct B site. For the two-sublattice encoding the kernel acquires the adjacent-bond cross term of section 3, at the same order and of the same footprint form; its weight is 5% on the warped chain and zero on the flat one. B–B shift bonds alone leave the zero mode unchanged to all orders (hβ = 0 forces β = 0, then h†α = 0 returns α = a); the small quadratic residual measured on isolated B bonds is the non-topological bulk background transmitted through the Lorentzian linewidth, identified by its η^2 scaling (constant to four digits across a 30× range of η, with the A-bond response η-independent as a zero-mode property must be). It is an artifact channel nine orders below signal at operating parameters and shrinks as the engineered linewidth narrows.

Curvature bias is a shift preamplifier, and the mechanism is the horizon. The $1/t_2^2$ gauge factor in the law grows wherever the bond profile is soft, and in the Rajbongshi–

Modak profile the horizon-end bonds vanish as N^−σ, so any fixed absolute shift becomes a large local tilt there: the warped-to-flat response ratio for the same near-edge feature scales as N^2σ — computed 42×, 85×, 174× and 369× at N = 50, 100, 200 and 400 for σ = 0.5. The factor is therefore a statement about the operating point, not a constant of the profile, and the correct operating variable is the local ratio $\tilde{w} = w/t_1$ at the probed edge: the same table shows $w_0/t_1(1)$ reaching 0.8 at N = 400, outside the perturbative regime, and the demonstration of section 5 sits at $w_0/t_1(1) \approx 0.4$. Raw-w factors are additionally convention-dependent (72× at N = 100 in the $t_2 = 1$ convention). A shift imager therefore wants a deliberately curved substrate.

And blindness beyond the footprint is the law's locality made literal: computed response to a warp bubble versus its position falls from 2×10^{-4} at five cells to exactly machine zero by thirty cells. The pixel is the footprint; a mid-chain feature is invisible to an edge element, and imaging extended shift structure requires the array.

5. Demonstration: a synthetic warp wall, with cited hardware

A 1+1 Alcubierre-style top-hat shift profile with smooth walls — flat inside, flat outside, all structure in two opposite-signed gradient walls — costs the mode nothing ($|E_0| \leq 10^{-18}$ with the bubble present) and produces a computed edge response of 1.4–1.8% in log-density for a two-percent shift amplitude parked near the element (the range reflects wall-geometry choices; both implementations agree within it). Against the companion paper's worked superconducting element — 3.3 aW signal power, demonstrated 30 zW/√Hz bolometry — that is 47–60 zW: **SNR ≈ 16–20 at 100 seconds of integration.** A static analog warp wall at two-percent shift amplitude is comfortably measurable with hardware already cited to demonstrated devices.

6. Array design rule: curvature-bias diversity

A single edge observable entangles the curvature and shift channels — the cross-responsivity $d \ln \rho/d\sigma$ itself runs with w. Computed response vectors at multiple element biases settle the disentangling scheme, and the convention-invariant statement is angular: a flat element paired with a warped element yields normalized response vectors separated by tens of degrees (38° and 21° in the two implementations — not the same computation: one evaluates both elements at a w = 0.02 bias, where the warped element's curvature responsivity has already collapsed from +0.20 to +0.03; the other at zero shift bias — with the same qualitative verdict), inverting the (σ, w^2) system stably; two elements differing only in shift bias are collinear to a fraction of a degree and separate nothing — obvious in hindsight from the law, since both share the same $1/t_2^2$ soft-region weighting. Both independent implementations reach the same qualitative verdict; raw condition numbers are deliberately not quoted, having been found normalization-dependent between conventions (a hazard the canonical variable $\tilde{w}$ removes if adopted end-to-end). **The array separates curvature from shift if and only if neighboring elements differ in curvature bias.** Fabrication cost: one lithographic profile choice per element, no new hardware.

7. Validation path

The photonic stage of the companion paper's ladder carries this result at no extra cost beyond mask design: a warped dimerized waveguide array with a current-like coupling (realizable as directional/auxiliary-ring coupling) exposes three quantitative predictions at room temperature — the quadratic responsivity in w with its closed-form coefficient, the exact evenness under sign reversal of the shift, and the single-site structure of the leaked amplitude, which is directly imageable in a photonic lattice. A null on any of the three falsifies the law before cryogenic investment. The superconducting stage inherits the companion paper's single differential measurement with one addition: a shift-biased element beside an unbiased one tests the preamplification factor and the channel separation in the same cooldown.

8. Honest scope

The dictionary is the standing debt, now partly paid by others. The identification of the current-like coupling with g_tx is by dispersion form — the odd-in-k identity term is the tilt of a 1+1 stationary metric — and the tilted-cone lattice literature [5] establishes position-dependent tilt as a horizon metric on a chain; what remains open is the extension of the curved-spacetime SSH derivation itself from static to stationary metrics, in the dimerized setting with an edge mode. That formal task remains open, and this paper's recommendation to it is concrete: define the lattice shift in local-hopping units from the start, since the convention-dependence of every raw-w statement was measured directly when two independent implementations disagreed by exactly the gauge factor the law prescribes. All results here are static; the response of the protected mode to a moving wall — formation and collapse analogs, the frontier the gravitational-waveform community has named — is untouched by these methods. Sign readout is excluded by symmetry, not difficulty, and belongs to a transport architecture. Array conclusions are two-element angular-separation statements; full-array error propagation under fabrication scatter is not yet computed. The law is stated exactly for A-sublattice shift and with its cross term for two-sublattice profiles; a general closed form for the cross term on an arbitrary warped profile is not given here, only its form on the flat chain and its magnitude on the warped one. And the entire result chain, while doubly implemented and cross-verified, has not been examined by anyone outside the two sessions that produced it.

9. Conclusion

The shift can be written into a topological chain without paying for it in protection: the symmetry changes hands rather than disappearing, the mode's energy stays pinned for arbitrary shift profiles, and the mode's shape becomes the sensor. Its law is one line — sensitivity is leaked probability, footprint-weighted — and it hands the analog-gravity community the same offer as the curvature law, extended to the metric class they actually care about: send the stationary metric, the law returns its lattice image and the response of every element placement, and a photonic array can falsify the whole construction for the cost of a mask.

Appendix A: Methods

Odd chains, 201 sites (100 cells), exact diagonalization and direct solve; power-law profile of the companion paper throughout; broadening $\eta = 10^{-3}$ as the engineered-linewidth proxy except in the η-scaling study (3×10^{-3} to 10^{-4}). Conventions: session one, $t_1 t_2 = 1$; session two, $t_2 = 1$; all convention-covariant claims verified in both. Raw-w responsivities agree between the conventions by the $1/t_2^2$ gauge factor only in the small-shift window $w \lesssim 0.02$ (ratio 2.0–2.1); at larger w the two conventions diverge nonperturbatively (10× by $w = 0.08$, with the curvature cross-talk changing sign), because a fixed absolute w is a different physical shift in the two conventions. This is the practical case for the canonical variable $\tilde{w}$: in local-hopping units the law is convention-free. Scripts (persisted with this draft): tilt_cost.py, tilt_cost2.py, bubble.py, reciprocal.py, derive_check.py, cross_term_check.py (session one); verify_return.py, kernel_law.py, law_check.py, derive_verify.py (session two).

Appendix B: Two-session verification protocol

Every result in this paper passed through two AI sessions with no shared code (two sessions of one model under one director, which is reimplementation, not independent review — the cold review in the repository's review/ directory is the first outside reading): claims were produced and computed in one session, reproduced from scratch in an independent implementation with a different hopping convention in the other, with pre-registered expectations where the protocol had matured. The record includes one productively falsified pre-registration (a two-point kernel conjecture, refuted by the exact diagonality later derived), one convention discrepancy opened in one session and

closed in the other by the disputed law's own gauge factor, and one corollary error (a quartic-order attribution) caught by the counterpart session's verification and resolved by an η-scaling measurement that identified the true mechanism. Tears are recorded, not smoothed; they are the evidence the checking was real.

References

[1] Companion draft: *A Topological Tidal-Curvature Imager* (B. Watson, 2026, unpublished; carries the full citation list for the SSH model, the curved-spacetime construction, qubit-probe prior art, superconducting platform parameters, and bolometry). [2] P. Rajbongshi and R. Modak, *Topological properties of curved spacetime extended Su-Schrieffer-Heeger model*, J. Phys.: Condens. Matter, DOI 10.1088/1361-648X/ae8635 (2026); arXiv:2504.16200. [3] Warp-metric physicality auditing: arXiv:2404.03095; Class. Quantum Grav. 41, 095009 (2024). *[verify before submission]* [4] Warp collapse waveforms: Clough, Dietrich, Khan, arXiv:2406.02466; Open J. Astrophys. 7 (2024). *[verify before submission]* [5] V. Könye, C. Morice, D. Chernyavsky, A. G. Moghaddam, J. van den Brink, and J. van Wezel, *Horizon physics of quasi-one-dimensional tilted Weyl cones on a lattice*, Phys. Rev. Research **4**, 033237 (2022); arXiv:2206.04138.